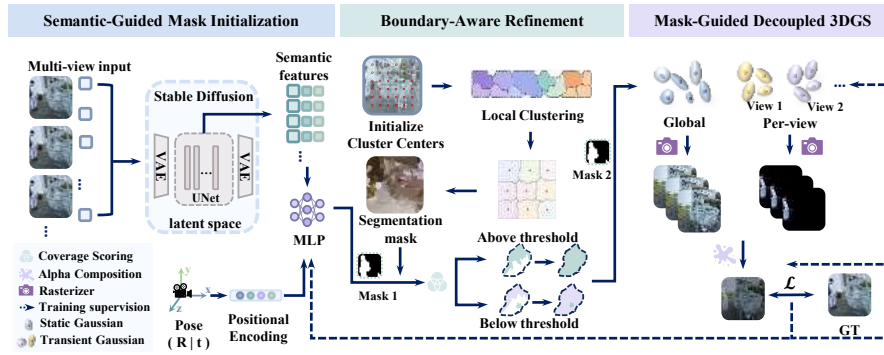


Graphical Abstract

TD-DGS: A Boundary-Aware Transient-Decoupled 3D Gaussian Splatting

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Abstract

Transient object removal aims to eliminate dynamic distractions from multi-view or multi-temporal inputs, thereby enabling stable and high-fidelity 3D scene reconstruction. However, existing 3D Gaussian Splatting (3DGS)-based approaches typically assume static scenes with consistent geometry and appearance, making them vulnerable to transient disturbances commonly found in real-world environments. These limitations manifest in three main aspects: (1) transient objects disrupt multi-view geometric consistency; (2) naive removal strategies often introduce visual artifacts and structural holes; and (3) conventional semantic masks often suffer from boundary noise and fragmented delineation when applied at the pixel level, especially under complex textures and illumination variations. To overcome these challenges, we introduce Transiently Decoupled 3D Gaussian Splatting (TD-3DGS). By explicitly decoupling the background from interfering elements, we enable their separate rendering, effectively eliminating distractors while preventing post-removal artifacts and holes. Furthermore, we employ a segmentation algorithm to address the problem of imprecise boundary detec-

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tion of interfering objects by semantic masks. Our approach achieves a 2.3% increase in Peak Signal-to-Noise Ratio (PSNR) compared to existing approaches, resulting in a significant improvement in reconstruction robustness and visual fidelity.

Keywords: Transient Distractors, 3D Gaussian Splatting, 3D Reconstruction, Decoupled Modeling, Computer Vision

1. Introduction

Transient-aware multi-temporal 3D reconstruction aims to recover clean and stable static 3D scenes from multi-view images captured at different time points. However, real-world image collections often contain transient disturbances, including dynamic objects such as pedestrians and vehicles, as well as temporary environmental variations such as lighting changes and shadows. These disturbances violate multi-view consistency and may introduce severe geometric artifacts, missing regions, or structural distortions. Therefore, robustly eliminating transient disturbances is essential for reliable 3D reconstruction in real-world applications such as E-commerce [1], autonomous driving [2], augmented reality (AR) [3], virtual reality (VR) [4], and biomedical imaging [5][6][7].

Inadequate removal of transient disturbances may introduce significant geometric artifacts, missing regions, and structural deformations, consequently impairing downstream tasks such as path planning, scene understanding, and human-computer interaction. To address these challenges, numerous methods have been proposed to improve the robustness of 3D reconstruction against transient disturbances. As shown in Fig. 1 (a), traditional approaches mainly rely on frame-to-frame photometric differences or multi-view geometric consistency to detect occluded regions. Recent methods further introduce regularization or guidance strategies, such as Occlusion Annealing Regularization (OAR) in DeclutterNeRF [8] and depth-guided mechanisms in Robust3DGS [9], to mitigate artifacts and holes. Other approaches, including WildGaussians [10] and T-3DGS [11], exploit visual features [12] or unsupervised schemes for occlusion

classification. However, these methods still show limited adaptability to static image collections captured at different time instances. In particular, they often struggle to distinguish transient occlusions from complex textures or illumination variations, resulting in multi-view inconsistencies, visual artifacts, and holes that degrade reconstruction quality.

Our work is inspired by the GaussianImage framework [13], which represents images as collections of 2D Gaussian distributions. This representation introduces a novel paradigm for image modeling and compression, enabling the modeling of distinct transient elements from multiple viewpoints. Furthermore, insights from the ECID [12] study demonstrate that diffusion models (e.g., Stable Diffusion [14]) are capable of implicitly learn cross-image semantic correspondences without explicit supervision. Remarkably, although these models are not explicitly trained for image matching tasks, they exhibit a strong ability to extract semantically consistent features across diverse visual inputs.

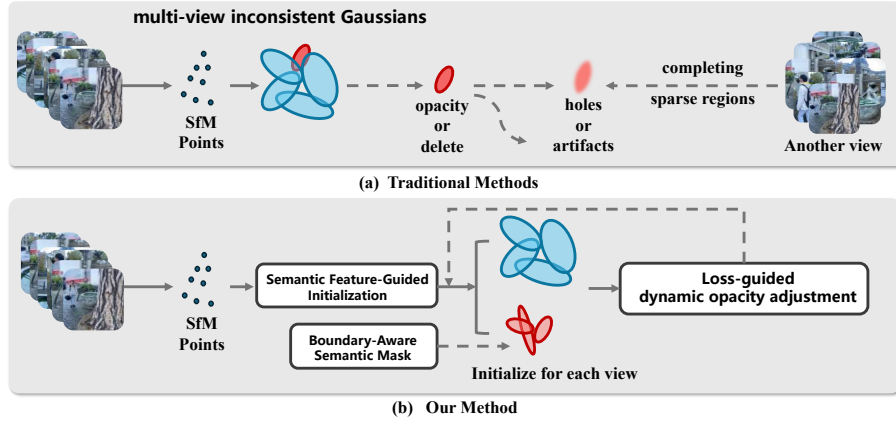


Figure 1: **Comparison with traditional methods.** (a) Traditional methods suppress inconsistent Gaussians by reducing opacity or directly removing points, which often results in holes and artifacts. (b) Our method decouples background and transient objects using semantic masks for separate reconstruction. Alpha blending with regularization dynamically adjusts opacities, while segmentation refinement enhances boundary consistency by aggregating mask responses within perceptually coherent regions. and minimizes side effects on background reconstruction.

Our TDR framework consists of two primary modules. First, the Semantic-Guided

Mask Initialization (SFMI) module leverages semantic features from the denoising process of a diffusion model to generate initial transient masks, providing more reliable initialization than residual- or decomposition-based masking methods such as DeGauss and DeSplat. Second, the Boundary-Aware Refinement (BAR) strategy incorporates segmentation results to refine mask boundaries and improve distractor localization. Unlike HybridGS, which uses image-specific 2D Gaussians for transient components, our method maintains a unified 3D Gaussian representation for both static and transient components while enabling view-adaptive transient decomposition. The overall pipeline is shown in Fig. 1 (b).

Experimental results on the On-the-go and Distractor datasets demonstrate that our method effectively removes transient objects while avoiding artifacts and holes, achieving accurate identification of dynamic distractors. Quantitatively, it achieves improvements of over 2.3% in Peak Signal-to-Noise Ratio (PSNR) and 2.1% in Structural Similarity Index Measure (SSIM) compared to existing methods.

The main contributions of our approach are as follows:

- Semantic-guided transient recognition: Leverages semantic features from diffusion models to learn accurate masks, enhancing dynamic occluder detection in complex scenes.
- Decoupled reconstruction framework: Separately models static backgrounds and transient objects, enabling robust and high-fidelity 3D reconstruction under transient occlusions.
- Segmentation-aware boundary refinement: Refines occlusion boundaries using segmentation results, improving geometric consistency and boundary precision.

The remainder of this paper is organized as follows: Section 2 reviews related work on reconstruction methods concerning transient removal, boundary refinement, and multi-view consistency, as well as reconstruction robustness in complex environments.

Section 3 outlines our approach. Section 4 presents an analysis of the experimental results. Finally, Section 5 concludes the paper.

2. Related Work

This chapter reviews the literature on transient distractor identification, separation, and consistent modeling in real-world 3D reconstruction. We first introduce methods for transient perception and modeling, with a particular focus on transient identification strategies under the NeRF and 3DGS frameworks. We then further discuss studies on transient removal, boundary refinement, and multi-view consistent reconstruction, and summarize the strengths, limitations, and implications of existing methods for the present work.

2.1. Transient Identification and Modeling

Identifying and modeling transient distractors in a scene is a prerequisite for robust reconstruction. Conventional Neural Radiance Fields (NeRF) [15] are constrained by the static-scene assumption and therefore tend to produce ghosting artifacts when processing videos containing transient objects. To alleviate this issue, researchers have proposed various scene decomposition and joint optimization strategies. NeRF-W [16] first introduced appearance embeddings to model transient occlusions in unconstrained image collections. Subsequently, RobustNeRF [17] adopted robust loss functions to model distractors as photometric outliers; NeRF-MS [18] explored scale-aware feature fusion strategies to mitigate boundary artifacts; NeRF On-the-go [19] further introduced uncertainty modeling to automatically localize distractors in complex environments; and NeRF-HuGS [20] combined sparse point cloud cues from SfM with hand-crafted heuristic rules to identify transient regions. In addition, D²NeRF[21] introduced two mutually independent neural radiance fields through self-supervised learning to decouple dynamic objects from static backgrounds, thereby effectively alleviating overfitting and occlusion issues. For complex urban scenes, RoDUS [22]

further adopted disentangled NeRF models together with a robust-kernel-based initialization strategy to achieve efficient decomposition of static and dynamic elements, while SUDS [23] combined neural radiance fields with static, dynamic, and far-field hash tables to realize unsupervised instance-level decomposition.

Compared with implicit representations, 3D Gaussian Splatting (3DGS) [24], owing to its explicit attributes and real-time rasterization capability, provides a more direct solution for filtering transient distractors. To enable distractor-free novel view synthesis, Robust3DGS [9] uses image residuals and a pretrained segmentation network to generate masks, and identifies and ignores distractors during optimization. SpotLessSplats [25], in contrast, introduces pretrained features and a robust optimization mechanism to encourage the model to automatically suppress distractor regions. T-3DGS filters transient objects through a bidirectional tracking module. RoGSplat mainly focuses on the independent regression and separation of human Gaussian parameters under sparse-view settings.

For large-scale urban environments, StreetGauss [26] and its follow-up work StreetGauss2 [27] specifically designed explicit motion separation strategies for dynamic elements such as vehicles and pedestrians. At the semantic perception level, EntityNeRF [28] achieves entity-level Thing/Stuff decomposition; NI-SGCN [29] models interactions among dynamic objects using a scene graph convolutional network; and MESC-3D [30] improves the purity of feature disentanglement by mining semantic cues.

Although existing methods have made significant progress in transient identification, in real-world unconstrained scenes they still suffer, after separation, from inaccurate transient region identification as well as missed detections and false positives for unseen object categories, due to insufficient exploitation of latent scene features. Moreover, heuristic pseudo-label construction or directly rendering foreground branches from photometric residuals may easily introduce noisy transient assignments in uncon-

strained scenes, especially for non-semantic disturbances such as shadows, reflections, and view-dependent artifacts.

2.2. *Transient Removal and Consistent Reconstruction*

After masks are obtained, achieving physically meaningful complete separation while preserving geometric consistency across multiple views is critical for high-quality reconstruction [31, 32]. D²NeRF [21] handles dynamic interference in monocular videos through self-supervised disentangled radiance fields. To address appearance variations in unconstrained environments, WildGaussians [10] and Wild-GS [33] introduce robust appearance modeling modules; WE-GS [?] proposes a residual-based spherical harmonics coefficient transfer strategy to alleviate photometric inconsistency. When processing multi-temporal satellite imagery, SkySplat [34] alleviates holes under sparse observations via a cross-view self-consistency module.

In terms of boundary processing and disentangled rendering, the static-dynamic disentanglement paradigm based on 3DGS has also achieved substantial progress. DeSplat [35] physically separates Gaussian primitives via a volume rendering mechanism; DeGauss [36] adopts a dual-Gaussian-set paradigm and combines it with probabilistic masks to complementarily optimize background and foreground. However, these methods often rely on foreground-background decomposition via foreground rendering or photometric residual assignment, implicitly assuming that transient disturbances are separable foreground objects. This assumption is unreliable in unconstrained scenes, where shadows, reflections, glare, and view-dependent appearance changes are often coupled with background surfaces, leading to inaccurate transient separation and unstable mask estimation.

RobustSplat [37] introduces a delayed densification strategy to disentangle transient interference; HybridGS [38] proposes a hybrid representation architecture that uses 2D Gaussians to represent transient objects and 3D Gaussians to represent static scenes, while enhancing their separability through viewpoint consistency. For sparse-

view settings, GS-W [39] and DepthSplat [40] introduce diffusion-based geometric priors and robust depth priors, respectively; SPAr3D [41] employs a point diffusion model to generate sparse 3D point clouds to guide mesh recovery; and SPIn-NeRF [42] combines multi-view segmentation with perceptual inpainting techniques to handle holes after separation.

Overall, existing methods improve reconstruction consistency to a certain extent through spatial constraints, but they still face issues such as imperfect alignment between mask boundaries and true object contours, as well as rendering ambiguities induced by global modeling.

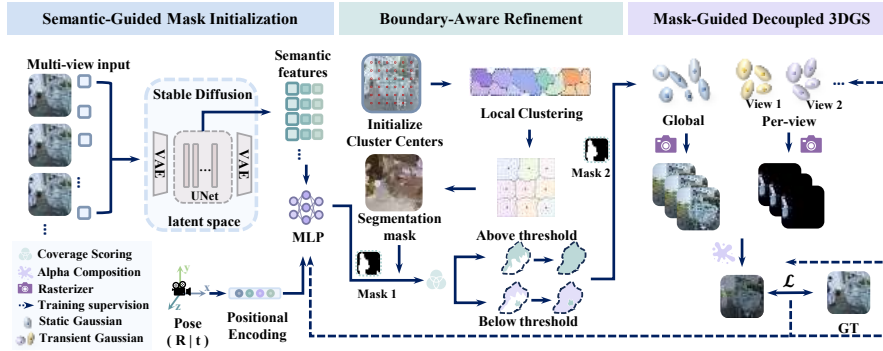


Figure 2: **An overview of our approach.** We first extract intermediate semantic feature representations from the latent space of the diffusion model, and train an MLP to produce an initial mask, M_1 . utilizing the segmentation outcome of the input image, we refine the boundary of M_1 to obtain an optimized mask M_2 . Finally, we initialize two sets of Gaussian points to represent the static background and distractor regions, respectively. The background Gaussians are initialized following the same strategy as in 3D Gaussian Splatting, while the distractor Gaussians are initialized per view using the refined mask M_2 . The background and distractors are rendered separately, and the final loss is computed and optimized after alpha blending their rendered results.

3. Our Approach

In this section, we detail our proposed framework, which leverages semantic guidance and spatial coherence to robustly reconstruct static scenes in the presence of transient distractors. Specifically, Section 3.1 introduces the semantic-guided mask initialization alongside a decoupled 3D Gaussian Splatting formulation, while Section 3.2

presents the boundary-aware refinement module designed to enforce region-level mask consistency, as illustrated in Fig. 2.

3.1. Semantic-Guided Mask Initialization

To accurately identify transient distractors, we design a semantic mask prediction MLP, which learns to generate an initial mask by jointly taking as input the semantic features extracted from the diffusion model and the positional encoding. We use the rendering residual as the sole self-supervisory signal, from which high-confidence pseudo-labels are constructed to guide the MLP to generate accurate transient object masks.

We first extract semantic features $s(p) \in \mathbb{R}^C$ from the latent space of a pre-trained diffusion model, and concatenate them with the positional encoding $\varphi(p) \in \mathbb{R}^K$ at pixel p , which is derived from the camera pose.

The input to the MLP is defined as $f(p) = [s(p), \varphi(p)]$. The MLP predicts the probability that pixel p belongs to the transient region as: $m(p) = \sigma(\text{MLP}(f(p)))$, where $\sigma(\cdot)$ denotes the Sigmoid function. Since $m(p)$ is a continuous probability value, we further binarize it to obtain the initial mask M_1 .

To train this MLP, we use the rendering residual $E(p) = |I(p) - \hat{I}(p)|$, as the sole supervisory signal, and generate pseudo-labels based on the quantiles of the residual:

$$\rho_l = \text{Quantile}(E(p), \tau_l), \quad \rho_u = \text{Quantile}(E(p), \tau_u). \quad (1)$$

Here, $\tau_l = 0.5$ and $\tau_u = 0.9$. Pixels whose residuals are greater than ρ_u are treated as transient object regions ($\mathcal{U}$, labeled as 1), while pixels whose residuals are smaller than ρ_l are treated as background regions ($\mathcal{L}$, labeled as 0).

We adopt the Hinge loss to supervise these high-confidence regions. This objective enforces the mask values to approach 1 in transient regions and 0 in background

regions:

$$\mathcal{L}_{\text{hinge}} = \frac{1}{|\mathcal{L}|} \sum_{p \in \mathcal{L}} \max(0, m(p)) + \frac{1}{|\mathcal{U}|} \sum_{p \in \mathcal{U}} \max(0, 1 - m(p)). \quad (2)$$

To encourage pixels with similar semantic features to obtain consistent transient responses, and to suppress local oscillations in mask prediction, we introduce a Lipschitz regularization constraint on the semantic mask generation MLP. This constraint is used to limit the magnitude of local variations in the network mapping, so that nearby pixels in the input feature space remain smooth and consistent in the output mask probabilities: $\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{Lip}}(\theta)$, where θ denotes the learnable parameters of the semantic mask generation MLP, and $\mathcal{L}_{\text{Lip}}$ denotes the Lipschitz regularization term for the smoothness of the MLP mapping. The final training objective for semantic mask generation is defined as:

$$\mathcal{L}_{\text{mask}} = \mathcal{L}_{\text{hinge}} + \lambda_r \mathcal{L}_{\text{reg}}. \quad (3)$$

Driven by the identification of transient regions, we adopt a decoupled 3D Gaussian Splatting framework to prevent transient components from corrupting the consistency of static geometry. The scene is modeled using a global static Gaussian set shared across all views, $\mathcal{G}_s = \{g_k^s\}_{k=1}^{N_s}$, and a transient Gaussian set bound to each individual view, $\mathcal{G}_d^{(i)} = \{g_j^{d,i}\}_{j=1}^{N_d^{(i)}}$.

Given the camera parameters π_i of view i and the static Gaussian set $\mathcal{G}_s$, we obtain the static rendered color $\mathbf{c}_s^{(i)}$. Similarly, the transient Gaussian set $\mathcal{G}_d^{(i)}$ produces the transient rendered color $\mathbf{c}_d^{(i)}$ and its accumulated opacity $\alpha_d^{(i)}$. The final composite color of view i is obtained through Alpha compositing:

$$\mathbf{c}^{(i)} = \mathbf{c}_d^{(i)} + (1 - \alpha_d^{(i)}) \mathbf{c}_s^{(i)}. \quad (4)$$

This compositing strategy ensures that when transient Gaussians are present, their colors are naturally overlaid onto the static background. When the transient Gaussians are absent or transparent, the rendering gracefully degenerates to a purely static scene,

thereby avoiding the holes or discontinuities caused by hard removal of transient regions.

The decoupled rendering output is supervised using the reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \sum_i \left((1 - \lambda_{\text{ssim}}) \|I_i - \mathbf{c}^{(i)}\|_1 + \lambda_{\text{ssim}} \mathcal{L}_{\text{D-SSIM}}(I_i, \mathbf{c}^{(i)}) \right). \quad (5)$$

To prevent the transient Gaussians from over-explaining the static background, while also avoiding holes in static regions caused by competition from the transient branch, we impose separate opacity constraints on the static Gaussians and the transient Gaussians. Specifically, the opacity of the transient Gaussians should remain sparse to reduce their erroneous explanation of the static background; in contrast, the opacity of the static Gaussians should accumulate sufficiently in background regions to stabilize the reconstruction of the static scene. Therefore, we define the opacity regularization terms as:

$$\mathcal{L}_{\text{static}} = \sum_i \|1 - \alpha_s^{(i)}\|_1, \quad \mathcal{L}_{\text{opacity}} = \sum_i \|\alpha_d^{(i)}\|_1, \quad (6)$$

where $\alpha_s^{(i)}$ and $\alpha_d^{(i)}$ denote the accumulated opacities of the static Gaussians and transient Gaussians in view i , respectively. The final optimization objective of the 3DGS stage is defined as:

$$\mathcal{L}_{\text{3DGS}} = \mathcal{L}_{\text{rec}} + \lambda_s \mathcal{L}_{\text{static}} + \lambda_d \mathcal{L}_{\text{opacity}}. \quad (7)$$

Here, λ_s and λ_d control the weights of the static opacity constraint and the transient opacity sparsity constraint, respectively.

The overall procedure is summarized in Algorithm 1.

3.2. Boundary-Aware Refinement

Although the initial mask M_1 can localize transient distractors, it usually suffers from jagged boundaries and local discontinuities. To address this issue, we introduce

Algorithm 1 TD-3DGS Training with Semantic-Guided Mask Initialization

Require: Images $\{I^{(i)}\}$, camera poses $\{\pi^{(i)}\}$, semantic features $\{s^{(i)}\}$
Ensure: Static Gaussians $\mathcal{G}_s$, transient Gaussians $\{\mathcal{G}_d^{(i)}\}$, masks $\{M^{(i)}\}$

- 1: **Initialize:** $\mathcal{G}_s$, $\{\mathcal{G}_d^{(i)}\}$, and mask MLP Φ_m
- 2: **for** each training iteration **do**
- 3: Sample a training view i
- 4: Extract semantic-position features $f(p) = [s(p), \varphi(p)]$
- 5: Compute residual map $E(p)$ from the current rendering
- 6: **Mask Initialization:** compute residual thresholds using Eq. (1)
- 7: Construct pseudo labels and predict mask $M^{(i)}$ with Φ_m
- 8: Update Φ_m by minimizing Eq. (3)
- 9: **Decoupled 3DGS:** optimize $\mathcal{G}_s$ and $\{\mathcal{G}_d^{(i)}\}$ guided by $M^{(i)}$
- 10: Obtain composite rendering $\mathbf{c}^{(i)}$ using Eq. (4)
- 11: Update $\mathcal{G}_s$ and $\{\mathcal{G}_d^{(i)}\}$ by minimizing Eq. (7)
- 12: **end for**
- 13: **return** $\mathcal{G}_s$, $\{\mathcal{G}_d^{(i)}\}$, $\{M_i\}$

a Boundary-Aware Refinement (BAR) module, which predicts the final accurate mask M_2 by suppressing jagged edges and local inconsistencies. The overall procedure is summarized in Algorithm 2.

We leverage the spatial coherence provided by superpixel segmentation to explicitly enforce region-level consistency on the initial mask. This intermediate step produces a more reliable boundary-aware mask. Specifically, during boundary refinement, we first refine the initial mask M_1 using the superpixel segmentation results to construct a superpixel-aware mask M_2 . Within each superpixel region, mask responses are aggregated to enhance consistency near boundaries, thereby producing a mask that better respects local image structures in a region-consistent manner and reduces spurious boundary fluctuations.

$$M_2(p) = \begin{cases} 1, & \text{if } p \in s_j \text{ and } \rho_j \geq 0.5, \\ M_1(p), & \text{otherwise.} \end{cases} \quad (8)$$

Here, ρ_j denotes the proportion of pixels labeled as transient objects within the

Algorithm 2 Boundary-Aware Mask Refinement with Superpixel Consistency

Require: Initial mask M_1 , mask probability $m(p)$, superpixel regions $\{s_j\}$, high-confidence regions $\mathcal{U}$ and $\mathcal{L}$

Ensure: Refined mask M_2

- 1: **Superpixel Segmentation:** partition the image into superpixel regions $\{s_j\}$
 - 2: **for** each superpixel region s_j **do**
 - 3: Compute transient proportion ρ_j according to Eq. (9)
 - 4: Refine M_1 to obtain M_2 according to Eq. (8)
 - 5: **end for**
 - 6: **Boundary-Aware Supervision:**
 - 7: Compute $\mathcal{L}_{\text{BCE}}$ on $\mathcal{U} \cup \mathcal{L}$ according to Eq. (10)
 - 8: Compute the total loss $\mathcal{L}_{\text{total}}$ according to Eq. (11)
 - 9: Update the framework by minimizing $\mathcal{L}_{\text{total}}$
 - 10: **return** M_2
-

superpixel region s_j :

$$\rho_j = \frac{1}{|s_j|} \sum_{p \in s_j} M_1(p). \quad (9)$$

where $M_1(p)$ is the binarized mask of $m(p)$.

If this proportion exceeds the threshold of 0.5, the mask within this region is enhanced, as shown in Fig. 3.

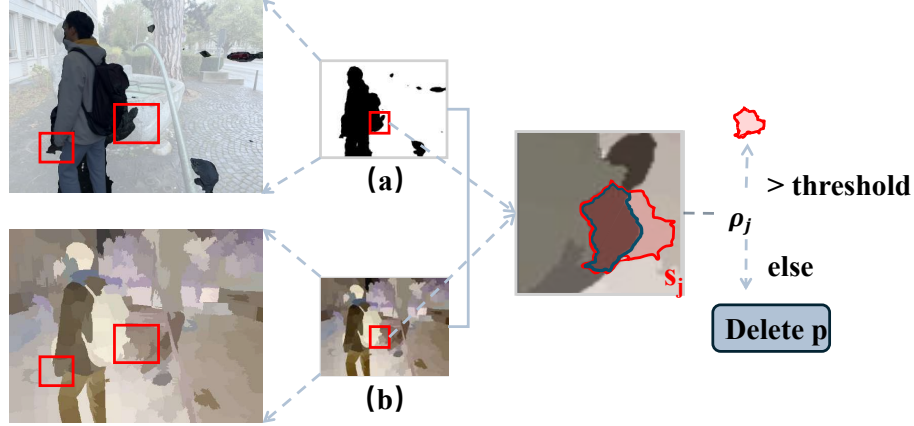


Figure 3: **Boundary-aware scoring mechanism with segmentation awareness.**(a) Initial mask M_1 . (b) Superpixel segmentation result. For each superpixel region s_j , if the proportion of pixels classified as transient objects ρ_j exceeds the threshold (0.5), the mask values within that region are enhanced by taking the maximum value in M_1 ; otherwise, the original mask M_1 is preserved.

To align the initial mask prediction with the superpixel-aware mask and avoid introducing unstable supervision in low-confidence regions, we adopt the binary cross-entropy loss only on the high-confidence transient region $\mathcal{U}$ and the high-confidence background region $\mathcal{L}$:

$$\mathcal{L}_{\text{BCE}} = \sum_{p \in \mathcal{U} \cup \mathcal{L}} \text{BCE}(m(p), M_2(p)). \quad (10)$$

Here, $\mathcal{U}$ and $\mathcal{L}$ denote the high-confidence transient region and high-confidence background region obtained from residual quantiles, respectively.

Combining the mask initialization and refinement stages, the overall end-to-end loss of the framework is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{3DGS}} + \lambda_1 \mathcal{L}_{\text{mask}} + \lambda_2 \mathcal{L}_{\text{BCE}}. \quad (11)$$

4. Experiments and Results

We assess the efficacy of our methodology and conduct a comparative analysis with alternative approaches using publicly accessible datasets.

Data Availability Statement: The two publicly available datasets used in this study, NeRF-on-the-go [19] and Distractor [17], can be accessed via their respective project websites.

4.1. Datasets and Evaluation Metricst

A systematic evaluation of the proposed method was conducted using two representative datasets: NeRF-on-the-go [19] and Distractor [17]. The NeRF-on-the-go [19] dataset comprises photo sequences of various indoor and outdoor scenes captured from randomly distributed viewpoints, featuring significant occlusion variations. It includes six test sequences, where the degree of occlusion across scenes ranges from 5% to 30%, making it well-suited for assessing robustness against occlusions.

In addition, we selected four standard benchmark scenes released by Distractor [17] (statue, Yoda, android, and crab) to evaluate the model’s capability in removing dynamic distractors. These sequences commonly exhibit the random appearance and disappearance of transient objects, and some scenes further contain continuously moving elements, such as floating balloons, posing a challenge for stable modeling under high uncertainty.

For quantitative evaluation, we employed three widely used image quality metrics: PSNR, SSIM, and Learned Perceptual Image Patch Similarity (LPIPS). PSNR and SSIM are used to measure pixel-level reconstruction fidelity and structural similarity, respectively, while LPIPS provides a perceptual assessment of visual quality.

4.2. *Implementation Details*

All experiments were conducted on a workstation with dual Intel i7-4790 CPUs (3.6 GHz), 64 GB RAM, and two NVIDIA RTX 3090 Ti GPUs. All methods were implemented in PyTorch and trained for 30,000 iterations using the Nerfstudio hyperparameter configuration. Static Gaussians were initialized from COLMAP sparse point clouds following the default Splatfacto strategy, optimized with Adam using the official 3DGS/Splatfacto learning rates, and trained with opacity reset for stable convergence.

For distractor Gaussians, we initialized 1,000 points per training view. Colors were represented as 3D RGB vectors without spherical harmonics and constrained to $[0, 1]$. Opacity reset was disabled to preserve view-specific transient information. Adam optimization used the following learning rates: position and opacity as default, rotation 0.01, scale 0.05, and RGB 0.025. Densification (ADC) was performed every 10 optimization steps per image with the same clone/split/cull policy as 3DGS. The rendering speed was comparable to Splatfacto, reaching approximately 100 FPS.

4.3. *Quantitative Comparisons*

In this section, we compare the proposed method with a series of related approaches, including the classical 3D Gaussian Splatting [24], the current state-of-the-art method

HybridGS [38], and representative techniques for handling dynamic occlusions such as RobustNeRF [17], NeRF On-the-go [19], WildGaussians [10], SpotLessSplats [25], Robust3DGS [9], and T-3DGS [11]. These methods explore transient object handling, occlusion removal, and high-fidelity reconstruction to varying extents, providing a comprehensive context for evaluating the performance and adaptability of our approach.

Table 1: Comparison on Nerf-on-the-go dataset.

Method	Mountain			Fountain			Corner			Patio			Spot			Patio-High		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
RobustNeRF [17]	17.54	0.49	0.38	15.65	0.31	0.57	23.04	0.76	0.24	20.39	0.71	0.25	20.65	0.62	0.39	20.54	0.57	0.36
NeRF On-the-go [19]	20.15	0.64	0.26	20.11	0.61	0.31	24.22	0.81	0.19	20.78	0.75	0.22	23.33	0.79	0.19	21.41	0.72	0.24
3DGS [24]	20.58	0.70	0.21	21.23	0.72	0.20	22.38	0.80	0.19	16.87	0.68	0.18	17.99	0.64	0.49	16.26	0.53	0.45
WildGaussians [10]	20.88	0.67	0.23	21.05	0.67	0.20	23.56	0.81	0.14	21.29	0.84	0.13	23.93	0.78	0.16	21.84	0.72	0.21
T-3DGS [11]	<u>22.49</u>	0.73	0.21	<u>22.84</u>	<u>0.79</u>	0.16	23.06	0.84	0.16	19.26	0.79	0.21	19.14	0.73	0.27	16.62	0.68	0.28
SpotLessSplats [25]	21.20	0.67	0.25	22.04	0.73	0.19	25.35	<u>0.87</u>	<u>0.10</u>	<u>22.60</u>	<u>0.85</u>	<u>0.07</u>	24.44	0.83	0.17	22.85	0.79	0.17
Robust3DGS [9]	16.06	0.42	0.58	15.69	0.36	0.56	19.02	0.72	0.36	17.95	0.76	0.26	15.58	0.49	0.51	18.83	0.67	0.31
Desplat [35]	20.75	<u>0.74</u>	<u>0.15</u>	22.31	0.78	<u>0.15</u>	<u>26.82</u>	0.86	0.13	18.09	0.77	0.12	<u>26.17</u>	<u>0.87</u>	<u>0.09</u>	23.22	<u>0.84</u>	<u>0.12</u>
HybridGS [38]	21.73	0.69	0.28	21.11	0.67	0.25	25.03	0.84	0.15	21.98	0.81	0.16	24.33	0.79	0.19	21.77	0.74	0.21
DeGauss [36]	22.31	0.73	0.16	22.40	0.76	0.13	25.94	0.87	0.11	22.58	0.85	0.08	26.59	0.86	0.08	<u>23.35</u>	0.07	<u>0.12</u>
Ours	23.42	0.83	0.10	23.97	0.86	0.13	27.84	0.89	0.08	23.22	0.89	0.05	26.82	0.90	0.11	24.34	0.89	0.11

Table 2: Comparison on distractor dataset.

Method	Statue			Android			Yoda			Crab1			Crab2		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
RobustNeRF [17]	20.60	0.76	0.15	23.28	0.75	0.13	29.78	0.82	0.15	32.22	0.94	0.06	32.47	0.88	0.11
NeRF On-the-go [19]	21.58	0.77	0.24	23.50	0.75	0.21	29.96	0.83	0.24	30.77	0.79	0.26	31.47	0.83	0.17
3DGS [24]	21.76	0.82	0.15	23.30	0.79	0.12	30.57	0.93	0.10	30.58	0.94	0.08	31.74	0.93	0.09
WildGaussians [10]	16.81	0.49	0.54	23.81	0.82	0.16	24.41	0.79	0.26	23.41	0.81	0.20	24.77	0.83	0.21
T-3DGS [11]	19.00	0.57	0.40	23.12	0.83	0.33	24.68	0.74	0.45	25.12	0.79	0.36	24.10	0.76	0.43
SpotLessSplats [25]	22.84	0.84	0.11	25.91	0.83	<u>0.07</u>	<u>35.72</u>	<u>0.95</u>	<u>0.07</u>	35.37	0.93	0.09	34.84	0.95	0.10
Robust3DGS [9]	15.29	0.49	0.46	23.69	0.84	0.13	24.29	0.83	0.25	25.09	0.90	0.17	24.05	0.85	0.23
Desplat [35]	<u>23.61</u>	<u>0.87</u>	0.10	24.73	0.84	<u>0.07</u>	35.69	0.95	0.10	36.10	<u>0.96</u>	0.06	35.01	0.93	0.08
HybridGS [38]	22.93	<u>0.87</u>	<u>0.11</u>	<u>25.15</u>	<u>0.85</u>	<u>0.07</u>	35.32	0.96	0.09	<u>36.31</u>	<u>0.96</u>	<u>0.05</u>	<u>35.17</u>	<u>0.95</u>	<u>0.07</u>
DeGaussian [36]	23.73	0.86	<u>0.11</u>	24.96	0.84	0.11	35.79	0.94	0.09	36.40	0.95	0.08	35.11	0.94	0.08
Ours	24.46	0.89	0.10	26.34	0.87	0.06	36.89	0.97	0.06	37.03	0.97	0.04	36.37	0.96	0.06

Tables 1 and 2 report the quantitative results on the On-the-go and RobustNeRF (Distractor) datasets, respectively. Our method achieves consistent improvements across PSNR, SSIM, and LPIPS, especially in heavily occluded scenes such as Spot and Patio-High. For scenes with fewer occlusions, our method remains comparable to existing approaches.



Figure 4: **Qualitative comparison on NeRF on-the-go dataset.** Multi-view comparison illustrating the presence of transient objects, marked by red bounding boxes.

4.4. Qualitative Comparisons

In this section, we present a visual comparison between the proposed method and a set of related approaches, including SpotLessSplats, Robust3DGS [17], T-3DGS [11], and Desplat [35]. These methods address transient objects, occlusion handling, and high-fidelity reconstruction to varying degrees, providing a solid basis for evaluating the performance advantages and adaptability of our approach.

Fig. 4 and Fig. 5 present the qualitative reconstruction results of our method on the Distractor and On-the-go datasets, respectively. The experimental results demonstrate that, compared to existing approaches, our method accurately identifies and eliminates transient distractors while preserving high-fidelity reconstruction quality. Notably, it exhibits superior performance in complex scenes characterized by dense occlusions.

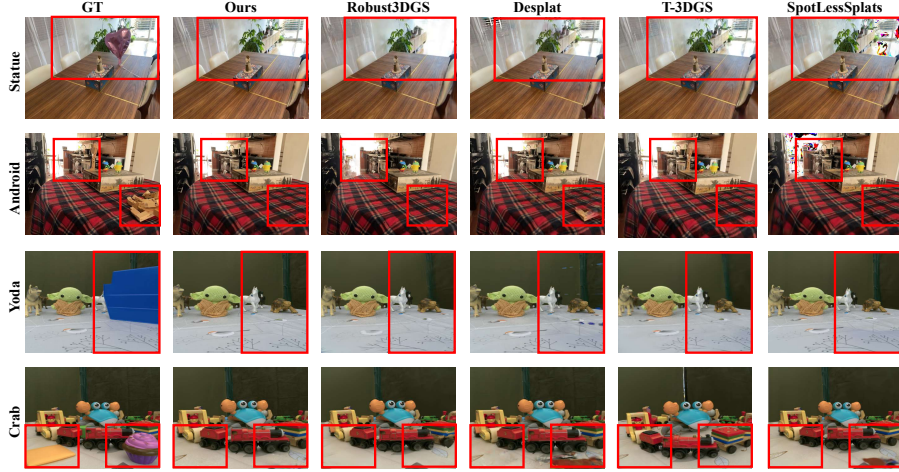


Figure 5: **Qualitative comparison on distractor dataset.** Multi-view comparison illustrating the presence of transient objects, marked by red bounding boxes.

4.5. Ablation Study

4.5.1. Ablation of Our Components

We adopt Splatfacto [43, 12] as the vanilla 3D Gaussian Splatting baseline [24], ensuring reproducible evaluation through the standardized Nerfstudio pipeline [12, 43]. Based on this baseline, we improve Gaussian point initialization, introduce an interference identification and removal module, and refine boundary accuracy in interference-affected regions.

We further conduct an ablation study across all scenes to evaluate the effectiveness of each component. As shown in Table 3, three configurations are compared: the baseline, only the Semantic-Guided Mask Initialization (SFMI) module, and the full model with both SFMI and Boundary-Aware Refinement (BAR). Each component consistently improves reconstruction quality, with the full model achieving the best overall performance. It should be noted that the BAR module operates on the masks produced by SFMI; thus, a configuration with only BAR is not feasible and is omitted from the study.

Table 3: **Ablation study results.** The upper part shows 6 scenes, and the lower part shows the remaining 5 scenes.

SFMI	BAR	Mountain			Fountain			Corner			Patio			Spot			Patio-high		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
✗	✗	20.58	0.70	0.22	21.23	0.73	0.21	22.38	0.80	0.20	16.87	0.68	0.19	17.99	0.65	0.49	16.26	0.54	0.45
✓	✗	21.20	0.68	0.26	22.04	0.73	0.20	25.35	0.88	0.10	22.60	0.85	0.06	24.44	0.84	0.17	22.85	0.79	0.18
✗	✓	22.02	0.83	0.18	22.84	0.83	0.17	25.06	0.85	0.16	19.26	0.79	0.22	19.14	0.74	0.28	16.62	0.68	0.29
✓	✓	23.70	0.83	0.10	23.97	0.86	0.15	27.84	0.89	0.08	23.22	0.89	0.07	26.75	0.87	0.11	24.34	0.89	0.11

SFMI	BAR	Statue			Android			Yoda			Crab(1)			Crab(2)		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
✗	✗	21.76	0.82	0.15	23.30	0.79	0.12	30.57	0.93	0.11	30.58	0.95	0.08	31.74	0.94	0.10
✓	✗	22.84	0.84	0.12	25.91	0.83	0.08	35.72	0.96	0.08	35.84	0.98	0.08	34.84	0.95	0.08
✗	✓	23.61	0.87	0.10	24.73	0.85	0.08	35.69	0.96	0.06	36.10	0.97	0.05	35.01	0.95	0.07
✓	✓	24.47	0.89	0.11	26.35	0.87	0.06	36.90	0.94	0.10	37.03	0.97	0.04	36.38	0.93	0.08

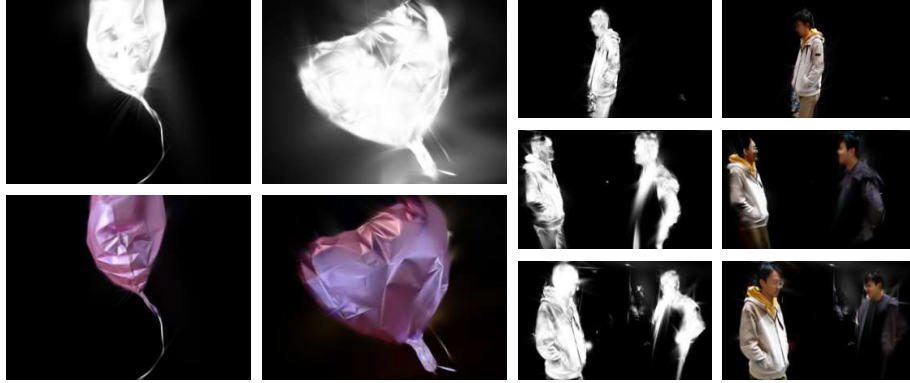


Figure 6: **Decoupled reconstruction.** Reconstruction Results of Transient Distractors from Different Viewpoints.

4.5.2. Semantic-Guided Mask Initialization

This module facilitates the separation and reconstruction of background Gaussians and transient objects during initialization. As shown in Fig. 6, the model successfully separates occluding components from the static background, resulting in enhanced accuracy of the reconstruction outcomes.

4.5.3. Boundary-Aware Refinement

Fig. 7 shows mask results from three viewpoints across two scenes. The initial mask M_1 is refined via our boundary optimization to produce the optimized mask M_2 . The optimized masks better align with transient object boundaries, demonstrating improved boundary coherence and reduced fragmentation compared to M_1 .

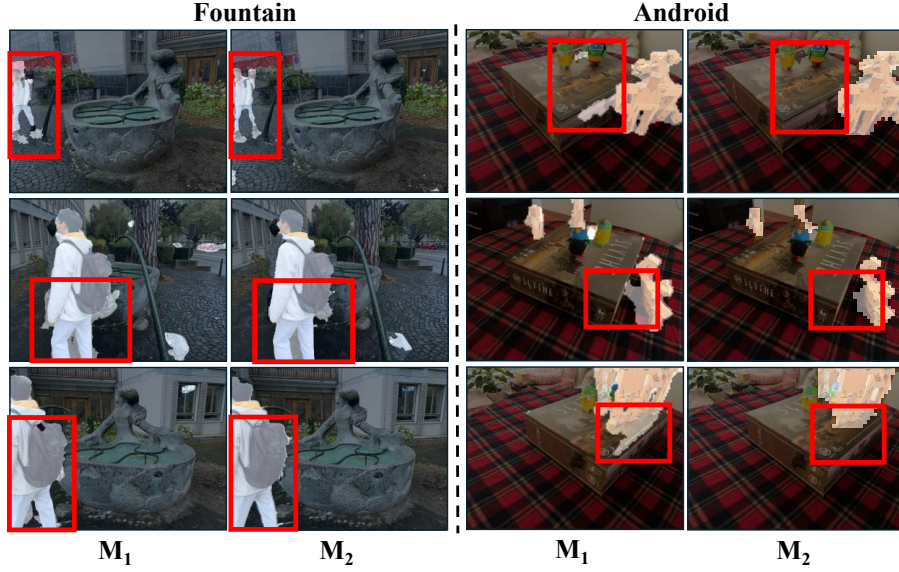


Figure 7: **Boundary-Aware Refinement.** We select three distinct viewpoints from two datasets for evaluation. M_1 refers to the initial semantic mask, and M_2 indicates the refined mask obtained through our optimization approach.

4.5.4. Hyperparameter Sensitivity Analysis

Opacity Regularization (λ_s, λ_d). These parameters modulate the decoupling between static backgrounds and transient interference. An undersized λ_s fails to adequately constrain static Gaussians, absorbing transient residuals and causing ghosting or floaters. Conversely, an oversized λ_s excessively penalizes static opacity, degrading background representation with missing details or local holes. For λ_d , a small value allows transient Gaussians to over-explain dynamic objects, destabilizing the separation. An oversized λ_d (e.g., 0.02) excessively penalizes transient opacity, forcing dynamic objects into the static background and creating ghosting artifacts. Setting $(\lambda_s, \lambda_d) = (0.01, 0.01)$ yields the optimal balance.

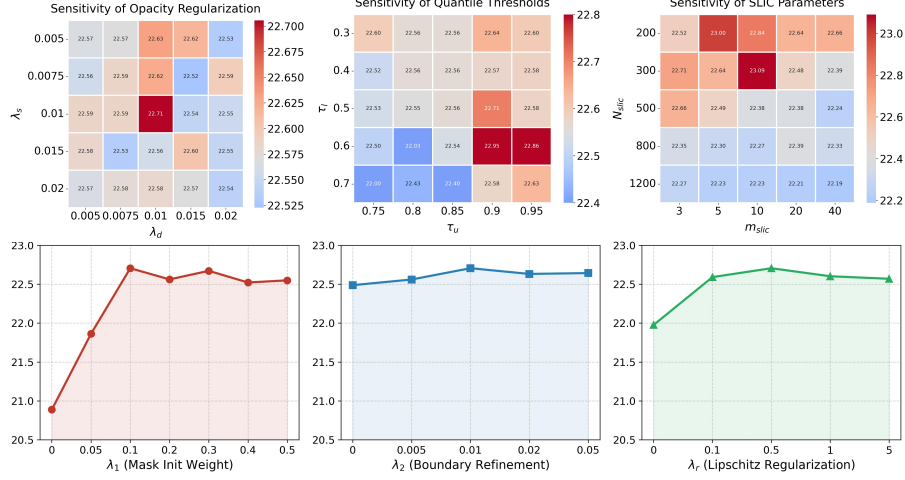


Figure 8: **Hyperparameter sensitivity analysis.** Sensitivity analysis of all hyperparameters, illustrating the average PSNR (dB) across all test scenes as a representative metric. The stable performance across moderate ranges demonstrates the overall robustness of our method, while the specific drops near zero validate the necessity of the proposed constraints.

Quantile Thresholds (τ_l, τ_u). These thresholds determine pseudo-label quality for self-supervision. An undersized τ_l (e.g., 0.3) yields insufficient background samples, hindering the mask network from learning complete background distributions. An oversized τ_l (0.7) introduces noisy, unstable boundaries into background labels. For transients, a low τ_u (0.7) generates overly permissive labels, misidentifying high-frequency static textures as transient. Conversely, an excessively high τ_u (0.95) produces a sparse sample set, slowing Mask MLP convergence and leaving transient masks incomplete. The (0.5, 0.9) configuration provides the most robust supervision.

Loss Weights ($\lambda_1, \lambda_2, \lambda_r$). These weights balance semantic-guided initialization, boundary-aware refinement, and Lipschitz regularization. An undersized λ_1 fails to leverage semantic priors, making early decoupling vulnerable to reconstruction errors. An oversized λ_1 over-relies on semantics, amplifying any detection errors from the semantic model. For λ_2 , an undersized value results in coarse, fragmented boundaries at object contours. An oversized λ_2 (0.05) overfits superpixel boundaries, erroneously propagating 2D segmentation errors to remove static regions. Finally, an undersized λ_r

lacks smoothness, causing jittery boundaries and fragmented masks; an oversized λ_r over-smooths, blurring fine-grained transient contours.

SLIC Parameters ($N_{\text{slic}}, m_{\text{slic}}$). These govern the boundary-aware refinement’s local consistency. An undersized N_{slic} produces large superpixels spanning both static and transient regions, exacerbating foreground-background mixing. An oversized N_{slic} over-segments the image, weakening local consistency and increasing vulnerability to texture noise. For compactness, a small m_{slic} forms irregular, fragmented regions in complex textures, while a large m_{slic} enforces rigid grids that fail to align with actual object contours.

Overall, the hyperparameter sensitivity analysis demonstrates that TD-3DGS exhibits strong stability across key parameters, despite variations in how these parameters mechanistically impact model performance. Ultimately, the experiments validate that our adopted parameter settings possess excellent generalizability and robustness, as illustrated in Fig. 8.

4.6. Discussion

To overcome the limitations of traditional transient removal methods, including imprecise masks and incomplete foreground-background separation, we introduce a semantic-guided Gaussian initialization mechanism. Specifically, a trained MLP generates robust occlusion masks based on semantic features, which are then used to initialize transient Gaussians for each view. We adopt a decoupled rendering strategy for background and transient components. By dynamically modulating the opacity of transient Gaussians, controllable transparency blending is achieved during synthesis, and transient elements to be progressively decoupled from background Gaussians throughout optimization, effectively suppressing artifacts and holes. Additionally, we design a boundary-aware mask refinement module that adaptively adjusts occlusion boundaries based on the mask proportion within superpixels, further enhancing boundary coherence and semantic consistency along object contours, without explicitly targeting

fine-grained pixel-level segmentation accuracy [44]. The collaborative effect of these modules significantly improves the accuracy of foreground-background separation in complex scenes, greatly enhancing the robustness and visual coherence of the final reconstruction.

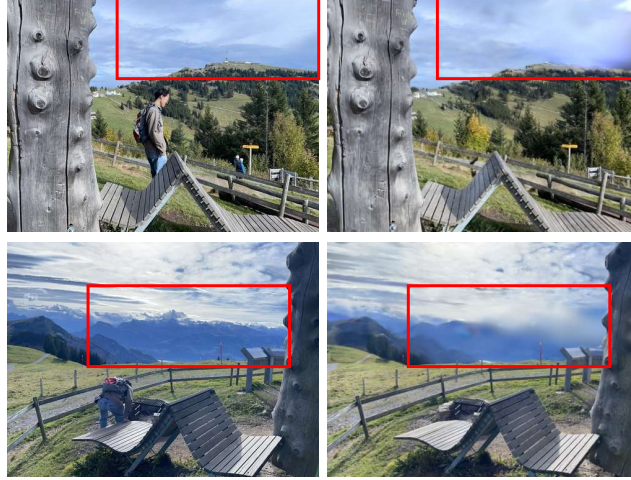


Figure 9: **Limitations of our method.** Comparison between the ground truth (left) and our reconstruction (right), illustrating areas where the proposed method falls short.

While our method achieves overall competitive performance, as shown in Fig. 9, the experiments highlight a limitation in reconstructing low-texture regions, as observed in the mountain scene where sky and cloud details are not well preserved. This issue primarily arises from the lack of distinctive gradients and geometric cues in low-texture regions, which hinders stable multi-view supervision and leads to uncertainty in the spatial distribution and opacity optimization of background Gaussians. Moreover, the absence of semantic-guided initialization for background points results in sparse distributions and incomplete boundary coverage.

To improve reconstruction in low-texture regions, future work will explore semantic-guided initialization of background Gaussians to provide better priors, and introduce multi-scale geometric regularization to enhance structural consistency during optimization.

5. Conclusion

In this paper, we introduce a boundary-aware decoupled modeling framework designed to improve occlusion detection robustness and boundary stability in the presence of transient distractors. within 3D Gaussian Splatting techniques for dynamic scene reconstruction. By explicitly distinguishing transient occluders from static background elements, our method effectively suppresses transient interference while significantly mitigating reconstruction artifacts and missing regions. Notwithstanding these advancements, our method continues to face challenges in accurately reconstructing fine-grained structures within low-texture regions, including areas such as skies and clouds. As part of future work, we plan to incorporate geometric priors and multi-scale regularization strategies to further improve reconstruction quality in such challenging regions.

CRedit authorship contribution statement

Tong Zhang: Writing – original draft, Writing – review & editing, Data curation, Software; **Rui Zhang:** Writing – review & editing; **Cuizhu Bao:** Writing – review & editing; **Mahmoud Hassaballah:** Writing – review & editing; **Aura Conci:** Writing – review & editing; **Abdulkadir Sengur:** Writing – review & editing; **Karen Egiazarian:** Writing – review & editing; **Yan Tian:** Supervision, Conceptualization, Methodology, Resources, Funding acquisition, Writing – review & editing;

Declaration of Competing Interest

All authors declare that they have no conflicts of interest with respect to the publication of this article.

Availability of Data and Materials

Data supporting the findings of this study are available from the corresponding author on a reasonable request.

Acknowledge

This work was supported by the Zhejiang Province Natural Science Foundation (No. LZ24F020001), the Opening Foundation of the Tongxiang Institute of General Artificial Intelligence (No. TAGI2-B-2024-0009), the program "Excellence initiative - research university" for the AGH University of Krakow, the ARTIQ project UMO-2021/01/2/ST6/00004 and ARTIQ/0004/2021, and Polish Ministry of Science and Higher Education funds assigned to the AGH University of Krakow, and the Open Project Program of State Key Laboratory of Virtual Reality Technology and Systems, Beihang University (No.VRLAB2026B13).

References

- [1] W. Shi, A. Yin, Y. Li, Y. Wen, Edge-guided 3D reconstruction from multi-view sketches and RGB images, *Pattern Recognition* 163 (2025) 111462.
- [2] Y. Cui, Z. Li, Z. Fang, Dynamic clustering transformer for lidar-based 3d object detection, *Pattern Recognition* 172 (2025) 112444.
- [3] L. Xia, X. Li, Z. Wang, H. Chen, X. Zhu, L. Fan, DMSAA-SLAM: RGB-D SLAM for dynamic scenes via diffusion self-attention, *Pattern Recognition* 179 (2026) 113576.
- [4] H. Zhao, Z. Guo, S. Qiao, Z. Gu, B. Zheng, J. Dong, H. Zheng, G2LFormer: Global-to-local token mixing transformer for blind image inpainting and beyond, *Pattern Recognition* 176 (2026) 113266.

- [5] Y. Tian, Y. Zhang, W.-G. Chen, D. Liu, H. Wang, H. Xu, J. Han, Y. Ge, 3D tooth instance segmentation learning objectness and affinity in point cloud, *ACM Transactions on Multimedia Computing, Communications, and Applications* 18 (4) (2022) 1–16.
- [6] Y. Tian, G. Jian, J. Wang, H. Chen, L. Pan, Z. Xu, J. Li, R. Wang, A revised approach to orthodontic treatment monitoring from oralscan video, *IEEE Journal of Biomedical and Health Informatics* 27 (12) (2023) 5827–5836.
- [7] Y. Tian, H. Fu, H. Wang, Y. Liu, Z. Xu, H. Chen, J. Li, R. Wang, RGB oralscan video-based orthodontic treatment monitoring, *Science China Information Sciences* 67 (1) (2024) 112107.
- [8] W. Liu, Z. Xiong, X. Li, N. Jacobs, DeclutterNeRF: Generative-free 3D scene recovery for occlusion removal, in: *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 380–390.
- [9] P. Ungermann, A. Ettenhofer, M. Nießner, B. Roessle, Robust 3D gaussian splatting for novel view synthesis in presence of distractors, in: *DAGM German Conference on Pattern Recognition*, 2024, pp. 153–167.
- [10] J. Kulhanek, S. Peng, Z. Kukulova, M. Pollefeys, T. Sattler, WildGaussians: 3D gaussian splatting in the wild, in: *Proceedings of the International Conference on Neural Information Processing Systems*, 2024, pp. 21271–21288.
- [11] A. Markin, V. Pryadilshchikov, A. Komarichev, R. Rakhimov, P. Wonka, E. Burnaev, T-3DGS: Removing transient objects for 3D scene reconstruction, in: *ArXiv Preprint ArXiv:2412.00155*, 2024.
- [12] L. Tang, M. Jia, Q. Wang, C. P. Phoo, B. Hariharan, Emergent Correspondence From Image Diffusions, in: *Conference on Neural Information Processing Systems*, 2023, pp. 1363–1389.

- [13] X. Zhang, X. Ge, T. Xu, D. He, Y. Wang, H. Qin, G. Lu, J. Geng, J. Zhang, Gaussianimage: 1000 FPS image representation and compression by 2D gaussian splatting, in: European Conference on Computer Vision, 2024, pp. 327–345.
- [14] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-Resolution Image Synthesis With Latent Diffusion Models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.
- [15] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, R. Ng, NeRf: Representing scenes as neural radiance fields for view synthesis, Communications of the ACM 65 (1) (2021) 99–106.
- [16] R. Martin-Brualla, N. Radwan, M. S. M. Sajjadi, J. T. Barron, A. Dosovitskiy, D. Duckworth, NeRF in the Wild: Neural Radiance Fields for Unconstrained Photo Collections, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021.
- [17] S. Sabour, S. Vora, D. Duckworth, I. Krasin, D. J. Fleet, A. Tagliasacchi, RobustNeRF: Ignoring distractors with robust losses, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 20626–20636.
- [18] P. Li, S. Wang, C. Yang, B. Liu, W. Qiu, H. Wang, NeRF-MS: Neural radiance fields with multi-sequence, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 18591–18600.
- [19] W. Ren, Z. Zhu, B. Sun, J. Chen, M. Pollefeys, S. Peng, Nerf On-the-Go: Exploiting uncertainty for distractor-free nerfs in the wild, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 8931–8940.
- [20] J. Chen, Y. Qin, L. Liu, J. Lu, G. Li, Nerf-Hugs: Improved neural radiance fields

- in non-static scenes using heuristics-guided segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.
- [21] T. Wu, F. Zhong, A. Tagliasacchi, F. Cole, C. Oztireli, D²nerf: Self-supervised decoupling of dynamic and static objects from a monocular video, *Advances in neural information processing systems* 35 (2022) 32653–32666.
 - [22] T.-A.-Q. Nguyen, L. Roldão, N. Piasco, M. Bennehar, D. Tsishkou, Rodus: Robust decomposition of static and dynamic elements in urban scenes, in: *European Conference on Computer Vision*, Springer, 2024, pp. 112–130.
 - [23] H. Turki, J. Y. Zhang, F. Ferroni, D. Ramanan, SUDS: Scalable urban dynamic scenes, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 12375–12385.
 - [24] B. Kerbl, G. Kopanas, T. Leimkühler, G. Drettakis, 3D gaussian splatting for real-time radiance field rendering, *ACM Transactions on Graphics* 42 (4) (2023) 139–1.
 - [25] S. Sabour, L. Goli, G. Kopanas, M. Matthews, D. Lagun, L. Guibas, A. Jacobson, D. Fleet, A. Tagliasacchi, SpotlessSplats: Ignoring distractors in 3D gaussian splatting, in: *ACM Transactions on Graphics*, Vol. 44, 2025, pp. 1–11.
 - [26] Y. Yan, H. Lin, C. Zhou, W. Wang, H. Sun, K. Zhan, X. Lang, X. Zhou, S. Peng, Street gaussians: Modeling dynamic urban scenes with gaussian splatting, in: *European Conference on Computer Vision*.
 - [27] R. Zhang, C. Li, C. Zhang, X. Liu, H. Yuan, Y. Li, X. Ji, G. H. Lee, Street gaussians without 3D object tracker, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 25722–25734.
 - [28] T. Otonari, S. Ikehata, K. Aizawa, Entity-nerf: Detecting and removing moving

- entities in urban scenes, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 20892–20901.
- [29] Z. Li, Q. Wu, J. Zhang, K. Zhang, J. Wang, Noise-injected spiking graph convolution for energy-efficient 3D point cloud denoising, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 39, 2025, pp. 18629–18637.
- [30] S. Li, Q. Cai, S. Kong, R. Tan, H. Tong, S. Qiu, Y. Jiang, Z. Liu, MESC-3D: Mining effective semantic cues for 3D reconstruction from a single image, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 16912–16921.
- [31] Y. Tian, P. Xue, W. Ding, M. Hassaballah, K. Egiazarian, A. Conci, A. Sengur, L. Rutkowski, DM-CFO: A diffusion model for compositional 3D tooth generation with collision-free optimization, *IEEE Transactions on Visualization and Computer Graphics* (2026) 1–13.
- [32] P. Xue, Y. Tian, Q. Song, Z. Wang, L. He, W. Ding, M. Hassaballah, K. Egiazarian, W.-f. Yang, L. Rutkowski, SVGS: Single-view to 3D object editing via Gaussian splatting, *ACM Transactions on Multimedia Computing Communications and Applications* (2026) 1–13.
- [33] J. Xu, Y. Mei, V. M. Patel, Wild-GS: Real-time novel view synthesis from unconstrained photo collections, *Advances in Neural Information Processing Systems* 37 (2024) 103334–103355.
- [34] X. Huang, X. Liu, Y. Wan, Z. Zheng, B. Zhang, M. Xiong, Y. Pei, Y. Zhang, SkySplat: Generalizable 3D Gaussian splatting from multi-temporal sparse satellite images, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 40, 2026, pp. 5158–5166.
- [35] Y. Wang, M. Klasson, M. Turkulainen, S. Wang, J. Kannala, A. Solin, Desplat:

- Decomposed Gaussian Splatting for Distractor-Free Rendering, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 722–732.
- [36] R. Wang, Q. Lohmeyer, M. Meboldt, S. Tang, Degauss: Dynamic-static decomposition with gaussian splatting for distractor-free 3D reconstruction, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025.
- [37] C. Fu, Y. Zhang, G. C. Kunbin Yao, Y. Xiong, C. Huang, S. Cui, X. Cao, Robust-Splat: Decoupling Densification and Dynamics for Transient-Free 3DGS, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 6294–6303.
- [38] J. Lin, J. Gu, L. Fan, B. Wu, Y. Lou, R. Chen, L. Liu, J. Ye, HybridGS: Decoupling Transients and Statics With 2D and 3D Gaussian Splatting, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 788–797.
- [39] D. Zhang, C. Wang, W. Wang, P. Li, M. Qin, H. Wang, Gaussian in the Wild: 3D Gaussian Splatting for Unconstrained Image Collections, in: European Conference on Computer Vision, 2024, pp. 341–359.
- [40] H. Xu, S. Peng, F. Wang, H. Blum, D. Barath, A. Geiger, M. Pollefeys, Depth-Splat: Connecting gaussian splatting and depth, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 16453–16463.
- [41] Z. Huang, M. Boss, A. Vasishta, J. M. Rehg, V. Jampani, Spar3D: Stable Point-Aware Reconstruction of 3D Objects From Single Images, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 16860–16870.
- [42] A. Mirzaei, T. Aumentado-Armstrong, K. G. Derpanis, J. Kelly, M. A. Brubaker, I. Gilitschenski, A. Levinshtein, SPIn-NeRF: Multiview segmentation and per-

ceptual inpainting with neural radiance fields, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 20669–20679.

- [43] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, T. Wang, A. Kristoffersen, J. Austin, K. Salahi, A. Ahuja, et al., Nerfstudio: A Modular Framework for Neural Radiance Field Development, in: ACM SIGGRAPH 2023 Conference Proceedings, 2023, pp. 1–12.
- [44] X. Zhao, J. Jiang, Y. Tian, L. Wu, Z. Xu, W.-f. Yang, Y. Zou, X. Wang, Innovative tooth segmentation using hierarchical features and bidirectional sequence modeling, *Pattern Recognition* 175 (2026) 113045.